Shuvayan Brahmachary

Bangalore, India | +91 93953 10075

b.shuvayan@gmail.com

github | linkedin

Summary

Sr. AI Engineer with 7 years of post-PhD experience across **India, Japan, and Germany**. Expert in **LLM-driven scientific reasoning, character language model, agentic AI frameworks, differentiable physics, reduced-order modeling, and optimization**. Proven track record leading high-impact projects and interns in collaboration with **NVIDIA, WPI, JAXA, KAIST, ISRO**, and others. Authored **16 journal papers** and several conference papers spanning AI, ML, CFD, and optimization, with strong focus on measurable business value and rNPV-linked outcomes.

Education

Indian Institute of Technology Guwahati (IITG) *2013–2019*

Ph.D. in Fluids and Thermal Engineering, Department of Mechanical Engineering

Gauhati University (GIMT) *2008–2012*

B.E. in Mechanical Engineering: University 2nd rank

Experience

Sr. AI Engineer, Databricks *Mar. 2026 – Present / India*

• Working in the AI-FDE team to delivery GenAI-based end-to-end solutions for customers deployed using Databricks tech stack.

SciML Researcher, Shell Plc *Sept. 2023 – Feb 2026 / India*

• **Project lead** towards the creation of LLM driven evolutionary optimizers LEO for complex single and multi-objective optimization. The framework was tested and benchmarked on industrial problems. The article was published in *Neurocomputing* (link) as lead author.

• **Project lead** on inhouse developed AI Agent GAMES for complex scientific simulation workflows by integrating GPT-4o, DeepSeek-R1, Llama-4-Scout, and O3-mini. Robust evaluation using LLM-as-a-judge and rule based evaluations. Projected impact: $14M rNPV.

• **Project lead** to setup robust guidelines for "Scientific Foundational Models" which led to review/position paper titled "*On Scientific Foundation Models: Rigorous Definitions, Key Applications, and a Survey*" (link) (Github Repo) in collaboration with Prof. Jagtap (WPI).

• Worked with in house geophysics experts to built spatio-temporal prediction model for Full-Wave Inversion (FWI) using Fourier Neural Operators (FNO). Accepted for *EAGE HPC Workshop 2025* (link) as lead author. Projected impact: $5M rNPV.

• Collaborated with CNOOC to created multi-objective optimisation framework for ML-based molecular discovery. Projected impact: **$19M rNPV**.

• Collaborated with fluid flow and reactor engineering team to developed ML-driven reactor-shape optimization framework for Catalytic Partial Oxidation (CPOX). Projected impact: **$8.5M rNPV**.

• Implemented hierarchical physics-informed NN for steady-state temperature prediction inside MCHEs using physics conserving neural network architecture; potential savings up to **$5M per LNG train**.

• Deployed DeepONet-based time-series model for MCHE cooldown fault detection in Nigeria NLG plant. Projected impact: **$19.5M rNPV**.

• Performed hierarchical clustering for semantic similarity in hard-hat image datasets using ResNet-50 embeddings.

Research Scientist, TUM *Jan. 2022 – Sept. 2023 / Germany*

• Built differentiable-physics ML framework for spatio-temporal flow-field prediction (*Physical Review E*, 2024) (link).

• Delivered 5 lectures (IN2298) on Advanced Deep Learning for Physics—optimization, convergence, and differentiable solvers. (link)

• Co-organized graduate seminar "Deep Learning in Physics" and mentored multiple master's projects. (link)

Research Scientist, Kyushu University *Sept. 2019 – Sept. 2021 / Japan*

• Developed surrogate-assisted evolutionary algorithms (SAEA) for aerodynamic shape optimization of high-speed vehicles. (link)

• Created ROM-assisted neural networks for scramjet intake and combustor; collaborations with JAXA and KAIST. (link)

• Led and mentored research interns, resulting in a featured publication in *Physics of Fluids*.

Independent Projects

Character Language Model

• Worked on training character language model (NanoGPT) on the Enwik8 test case using cloud HPC platform. The goal of this project was to reduce the bits per character (bpc) of the model (size 44M) to 1.1 (and lower) using novel innovations such as rotary positional embeddings, Kolmogorov-Arnold Networks, Sparse Attention, etc.

Stable Diffusion for text to Image

• Worked towards training a Stable Diffusion (SD) models on the KANJIDIC2 dataset keeping the text encoder frozen, so that the resulting trained (or fine tuned) model performs text-to-image tasks.

AI Scientist

• Worked on an Agentic AI-based framework called AI Scientist for novel manuscript writing using 'claude-3-5-sonnet-20240620' LLM. This repo puts the AI Scientist to test on an entirely new problem, i.e., Sparse Identification of Non-linear Dynamical systems (SINDy). (link)

MultiAgent Debate

• Worked on a multi-agent debate-engagement framework for mathematical problems. Worked on reproducing the number of agents vs accuracy plot in Du et al. 2024 . Similar results have also been obtained in Chan et al. 2023.

Training and Certifications

- Machine Learning in Production (deeplearning.ai) (link)
- Generative AI with Large Language Models (deeplearning.ai) (link)
- Large Language Model Agents (UC Berkeley, ongoing) (link)
- LangChain for LLM Application Development (deeplearning.ai) (link)
- Data-Driven Engineering Workshop, Isaac Newton Institute (2023) (link)
- ICME Stanford Summer Workshop 2021: ML, DL, and HPC (link)
- R&D Valuation and Technology Economics (Shell Plc)

Selected Publications

- Brahmachary, S., Thuerey, N.,*"Unsteady Cylinder Wakes from Arbitrary Bodies with Differentiable Physics-Assisted Neural Network"* (**Physical Review E**, 2024) Link
- Menon, S., Mondal, T., Brahmachary, S., Panda, A., Joshi, S, M., Kalyanaraman, K., Jagtap, A, D., *"On Scientific Foundation Models: Rigorous Definitions, Key Applications, and a Survey"* (**SSRN**, 2025, Link
- Brahmachary, S., Joshi, S., Panda, A., Konerapalli, K., Sagotra, A., Patel, H., Sharma, A., Jagtap, A., Kalyanaraman, K. "Large Language Model-Based Evolutionary Optimizer: Reasoning with Elitism" (**Neurocomputing**, 2024) Link

Others

- Invites talks at Paral-Intelligence seminar series, Worcester Polytechnic Institute WPI, Karlsruhe Institute of technology (KIT, Germany) and Space Transportation Systems Engineering Laboratory, Kyushu University (STSEL, Japan)
- Blogs on topics "When Kolmogorov-Arnold network meet PiNNs: Good, Bad and the Ugly Link", "Differentiable Physics for Unsteady Fluid Dynamics Link", "Principal component analysis in aerodynamic shape optimisation Link"
- Life member: Society for Shock Wave Research (SSWR) and Member of the Soft Computing Research Society (SCRS)